

# Roshan Taneja

Machine Learning · Geospatial Data Science · Avionics Systems

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## EDUCATION

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| <b>B.S. Electrical Engineering &amp; Computer Science (EECS) @ UC Berkeley</b> (Berkeley, CA)                                                                                                                                                                                                                        | 2025 – 2029 |
| ◇ Structure & Interpretation of Computer Programs · Data Structures & Algorithms · Linear Algebra · Discrete Mathematics & Probability Theory · Multivariable Calculus · Introduction to Circuits & Devices · Foundations of Signals, Dynamical Systems, & Information Processing · Introduction to Amateur Rocketry |             |
| <b>Spatial Research Certificate @ Harvard GIS Institute</b> (Cambridge, MA)                                                                                                                                                                                                                                          | Spring 2025 |
| <b>Intermediate Data Science @ UCLA</b> (Los Angeles, CA)                                                                                                                                                                                                                                                            | Summer 2024 |
| <b>Remote Sensing @ MIT Beaver Works</b> (Cambridge, MA)                                                                                                                                                                                                                                                             | Spring 2024 |
| <b>High School Diploma — GPA 3.98 · ACT 36 @ Sacred Heart Preparatory</b> (Atherton, CA)                                                                                                                                                                                                                             | 2021 – 2025 |

## PROJECTS & WORK EXPERIENCE

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| <b>Backend Engineer @ Statewide Database</b> (Berkeley, CA)                                                                                                                                                                                                                                                                                     | 2026 – Pres. |
| ◇ Built end-to-end voter-registration ingestion pipeline in <b>Dagster + DuckDB</b> , processing county-level fixture files through validation (Pydantic), translation, and loading with correct category/geography detection. Reduced manual validation overhead by automating schema checks across multi-format registration files.           |              |
| ◇ Deploying a standalone frontier LLM (Ollama) for communal use through the Lab Intranet.                                                                                                                                                                                                                                                       |              |
| <b>Avionics Engineer @ Cal Space Technologies and Rocketry (STAR)</b> (Berkeley, CA)                                                                                                                                                                                                                                                            | 2025 – Pres. |
| ◇ Designing avionics systems for high-power rockets (~30,000 ft apogee); custom PCBs via Altium Designer. Soldering and testing high-/low-voltage boards under strict mass and volume constraints; firmware in ESP32/C++.                                                                                                                       |              |
| <b>Research Intern — Center for Geographic Analysis @ Harvard University</b> (Cambridge, MA)                                                                                                                                                                                                                                                    | 2024 – 2025  |
| ◇ Collaborated with Prof. Wang (CGA) to develop <b>satellite ML models</b> identifying Maasai settlement patterns across <b>10,000+ image tiles</b> to optimise placement of rainwater harvesting units in Tanzania. Expanded model to new geographies; work underpins active humanitarian deployment.                                          |              |
| <b>Data Science Researcher @ Cost of Colonialism (COCO)</b> (Remote)                                                                                                                                                                                                                                                                            | 2024 – 2025  |
| ◇ Built <b>OCR + NLP semantic search</b> over 1,600s–1,800s Dutch East India Company shipping manifests; <b>embedding-based retrieval</b> with historian-facing query UI. Automated entity extraction across noisy archival scans, reducing manual review time from weeks to minutes.                                                           |              |
| <b>Youth President, Water Programs @ Maji Wells</b> (Losimingori, TZ)                                                                                                                                                                                                                                                                           | 2021 – 2025  |
| ◇ Led development of an <b>ML satellite image classifier</b> to locate Maasai communities for infrastructure planning. <b>Fundraised \$300,000</b> to deploy <b>500 rainwater harvesting solutions</b> directly serving <b>30,000 Maasai</b> across Tanzania — core work behind the 2024 UN SDG #6 Award and Children’s Peace Prize nomination. |              |

## PUBLICATIONS, PATENTS & CONFERENCES

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| 🏆 <b>NeurIPS 2024 — "Machine Learning for Social Impact" (1 of 4 Winners)</b>                              | 2024 |
| 14,000 attendees; selected from global submissions across all career stages.                               |      |
| 💡 <b>Provisional Patent: Object Detection in Satellite Imagery for Optimising Rainwater Unit Placement</b> | 2024 |
| US Patent Application 63/703,232 · Published in <i>National High School Journal of Science (NHSJS)</i> .   |      |
| 🎤 <b>Keynote Presenter — Machine Learning for Earth Observation (ML4EO) Conference</b>                     | 2024 |
| University of Exeter, UK · Sole high-school keynote speaker.                                               |      |
| 📖 <b>Publication: Impact of Rainwater Harvesting on 4,500+ Maasai</b>                                      | 2024 |
| <i>National High School Journal of Science (NHSJS)</i> .                                                   |      |
| 📄 <b>AICC @ NeurIPS — Poster Presentation</b>                                                              | 2025 |
| Copenhagen, Denmark.                                                                                       |      |

## TECHNICAL SKILLS

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| <b>Languages:</b> Python · C++ · Java · JavaScript / TypeScript · AppScript · SQL                                                    |
| <b>ML / AI:</b> YOLOvX · CNNs · RetinaNet · Transfer Learning · Object Detection · Embedding-based Retrieval · OCR + NLP             |
| <b>Geospatial:</b> ArcGIS Pro · Google Earth Engine · PostGIS · GeoPandas · AlphaEarth · MCDA · SOAR · mAP/F1 evaluation             |
| <b>Infrastructure:</b> Docker · Linux · DuckDB · Dagster · Pydantic · Full-Stack Web Development                                     |
| <b>Hardware:</b> Altium Designer · KiCad · ESP32 · PCB Design · CAD/CAM (Fusion 360, SolidWorks) · CNC · 3D Printing · Laser Cutting |